

Performance-Focused Design Exercise Packet

Part of ASME IDETC 2025 Workshop:

Adventures in Tradespace Exploration

17 August 2025

Group ID: 3000



Annotate, mark-up, and write on this page as needed.

Post-Activity Questions:

Describe the solution the group has selected – write in the value of the design variables.

19.2 in	92.3 in	76.1 in	8.6 in	915 11-12	4.29 = 1	45 in	2.90 in	0.773 in	0.18 in
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20

Do you think you could have found a better solution given more time?

Not really – given the 4 variables on the plots we believe we found the best solution for both teams.

What, in this round, stopped you from finding a better solution?

More variables could have contributed to finding a more refined solution. Also, the trade-offs for each option were clear – so we were able to deduce the best solution fairly easily.

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Round 2:

For this design exercise, you will be acting as a member of your assigned working group, and you will be provided cross-disciplinary performance space plots.

INSTRUCTIONS:

Working in your assigned working group, select a single solution using the performance space plots provided in the supplemental packet. You may annotate and mark the supplemental packet as you wish. You wish to maximize the vehicle's performance in your own objectives, and the other working group wishes to maximize their own objectives, but you must select a single vehicle from the 20 designs given in the supplemental packet. Write the selected design's ID in the space provided in the supplemental packet.

The Suspension and Driveline working group wishes to maximize the following objectives:

E. Top Speed

F. Off Road Ability

2, 5, 10, 3, 20

The Chassis and Cabin working group wishes to maximize the following objectives:

G. Occupant Protection

H. Maintainability

2, 8, 12, 9, 11

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Post-Activity Questions:

Describe the solution the group has selected – write in the value of the design variables.

110 in	892 in	65.5 in	20.0 in	441 lb-ft	2.15 ± 1	44 in	1660 Rev	0.979 in	1.48 in
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#2 Clear winner in most categories

Do you think you could have found a better solution given more time?

No – Based on the plots given, 2 is the best solution for all objectives

What, in this round, stopped you from finding a better solution?

More variables would have helped,
The vehicle statistics revealed more than
the plots did.

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Post-Activity Questions:

Describe the solution the group has selected – write in the value of the design variables

110 in	92 in	67.4 in	20.6 in	517 lb-ft	2.84 1	33 in	1210 12Wh	.432 in	Very heavy Armor
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#14

Do you think you could have found a better solution given more time?

maybe, we were crunched for
time

What, in this round, stopped you from finding a better solution?

Time + deciding the most important
factors

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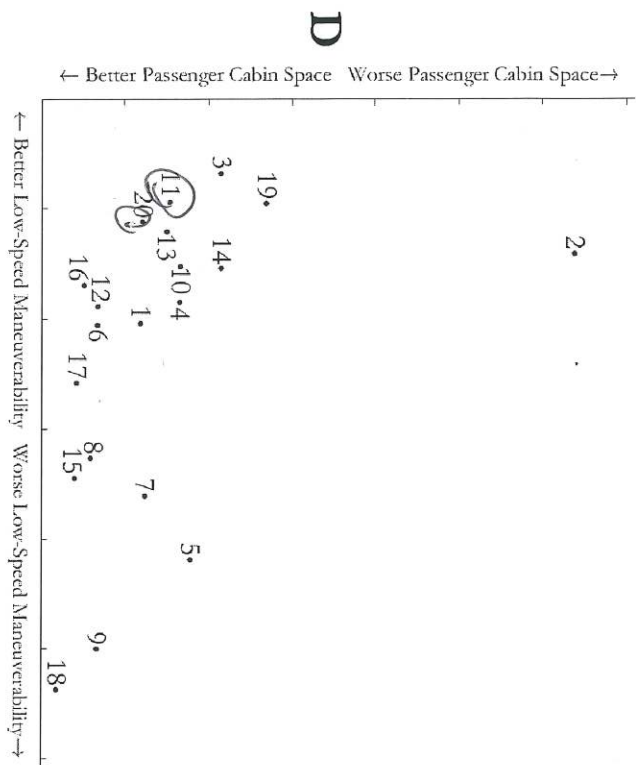
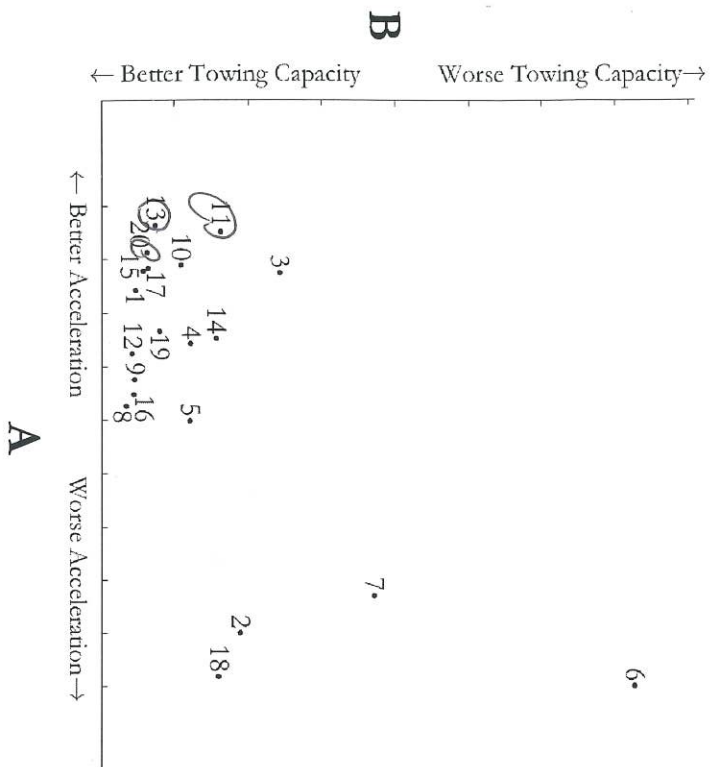
Vehicle Index

	Trackwidth	Wheelbase	Roof Height	Suspension Travel	Max Enging Torque	Driveline Gearing	Tire Diameter	Energy Capacity	Frame Rail Wall Thickness	Body Panel Thickness
1	196 in	80.2 in	79.3 in	22.9 in	728 lb-ft	5.87:1	46 in	1620 kWh	0.948 in	0.98 in
	Very Wide	Very Short	Tall Roof	Very Long Travel	Very High Output	Very Short Gearing	Very Large Tire	Very High Energy Capacity	Very Heavy-Duty Frame	Moderately Armored
2	110 in	80.1 in	65.5 in	20.8 in	421 lb-ft	2.15:1	49 in	1660 kWh	0.939 in	1.48 in
	Very Narrow	Very Short	Very Short Roof	Very Long Travel	Very Low Output	Very Long Gearing	Very Large Tire	Very High Energy Capacity	Very Heavy-Duty Frame	Heavily Armored
3	119 in	80.2 in	70.2 in	6.8 in	543 lb-ft	4.67:1	48 in	1560 kWh	0.285 in	0.18 in
	Very Narrow	Very Short	Short Roof	Very Short Travel	Moderate Output	Short Gearing	Very Large Tire	Very High Energy Capacity	Very Light-Duty Frame	Very Lightly Armored
4	181 in	81.2 in	69.2 in	9.7 in	655 lb-ft	2.37:1	32 in	1070 kWh	0.877 in	1.99 in
	Very Wide	Very Short	Short Roof	Short Travel	High Output	Very Long Gearing	Very Small Tire	Low Energy Capacity	Very Heavy-Duty Frame	Very Heavily Armored
5	192 in	85.6 in	65.3 in	21.1 in	684 lb-ft	2.61:1	45 in	1240 kWh	0.783 in	1.48 in
	Very Wide	Short	Very Short Roof	Very Long Travel	Very High Output	Very Long Gearing	Large Tire	Moderate Energy Capacity	Heavy-Duty Frame	Heavily Armored
6	178 in	81.1 in	79.1 in	6.8 in	409 lb-ft	2.53:1	47 in	1530 kWh	0.251 in	0.47 in
	Wide	Very Short	Tall Roof	Very Short Travel	Very Low Output	Very Long Gearing	Very Large Tire	Very High Energy Capacity	Very Light-Duty Frame	Very Lightly Armored
7	199 in	84.2 in	68.2 in	22.7 in	419 lb-ft	2.4:1	39 in	1520 kWh	0.331 in	1.74 in
	Very Wide	Short	Very Short Roof	Very Long Travel	Very Low Output	Very Long Gearing	Moderate Tire	High Energy Capacity	Very Light-Duty Frame	Very Heavily Armored
8	176 in	98.8 in	69.2 in	10.8 in	443 lb-ft	5.08:1	31 in	960 kWh	0.994 in	1.67 in
	Wide	Very Long	Short Roof	Short Travel	Very Low Output	Short Gearing	Very Small Tire	Very Low Energy Capacity	Very Heavy-Duty Frame	Very Heavily Armored
9	195 in	96.4 in	66.1 in	9.3 in	630 lb-ft	5.12:1	50 in	1610 kWh	0.741 in	0.17 in
	Very Wide	Very Long	Very Short Roof	Very Short Travel	High Output	Short Gearing	Very Large Tire	Very High Energy Capacity	Heavy-Duty Frame	Very Lightly Armored
10	134 in	83.1 in	80 in	18.1 in	706 lb-ft	5.95:1	49 in	1010 kWh	0.512 in	1.75 in
	Narrow	Very Short	Tall Roof	Long Travel	Very High Output	Very Short Gearing	Very Large Tire	Very Low Energy Capacity	Light-Duty Frame	Very Heavily Armored
11	107 in	87.6 in	78 in	8.1 in	750 lb-ft	2.93:1	30 in	880 kWh	0.335 in	1.97 in
	Very Narrow	Short	Tall Roof	Very Short Travel	Very High Output	Long Gearing	Very Small Tire	Very Low Energy Capacity	Very Light-Duty Frame	Very Heavily Armored
12	114 in	99.7 in	82 in	18.4 in	715 lb-ft	2.25:1	30 in	980 kWh	0.967 in	0.74 in
	Very Narrow	Very Long	Very Tall Roof	Long Travel	Very High Output	Very Long Gearing	Very Small Tire	Very Low Energy Capacity	Very Heavy-Duty Frame	Lightly Armored
13	107 in	98.2 in	66.6 in	23.1 in	512 lb-ft	5.32:1	30 in	1530 kWh	0.284 in	0.84 in
	Very Narrow	Very Long	Very Short Roof	Very Long Travel	Low Output	Very Short Gearing	Very Small Tire	High Energy Capacity	Very Light-Duty Frame	Lightly Armored
14	110 in	92 in	67.4 in	20.6 in	517 lb-ft	2.84:1	33 in	1210 kWh	0.432 in	1.82 in
	Very Narrow	Long	Very Short Roof	Very Long Travel	Low Output	Long Gearing	Very Small Tire	Moderate Energy Capacity	Light-Duty Frame	Very Heavily Armored

Vehicle Index

15	200 in	95.1 in	81.7 in	20.6 in	665 lb-ft	5.92:1	31 in	1460 kWh	0.313 in	1.67 in
	Very Wide	Long	Very Tall Roof	Very Long Travel	High Output	Very Short Gearing	Very Small Tire	High Energy Capacity	Very Light-Duty Frame	Very Heavily Armored
16	124 in	99.8 in	82.1 in	18.8 in	532 lb-ft	3.62:1	31 in	1570 kWh	0.782 in	0.33 in
	Narrow	Very Long	Very Tall Roof	Long Travel	Low Output	Moderately Gearing	Very Small Tire	Very High Energy Capacity	Heavy-Duty Frame	Very Lightly Armored
17	172 in	94 in	83.4 in	9.6 in	735 lb-ft	5.94:1	43 in	1500 kWh	0.313 in	0.78 in
	Wide	Long	Very Tall Roof	Very Short Travel	Very High Output	Very Short Gearing	Large Tire	High Energy Capacity	Very Light-Duty Frame	Lightly Armored
18	192.00527	98 in	82.4 in	6.5 in	442 lb-ft	3.23:1	35 in	1120 kWh	0.308 in	1.7 in
	Very Wide	Very Long	Very Tall Roof	Very Short Travel	Very Low Output	Long Gearing	Small Tire	Low Energy Capacity	Very Light-Duty Frame	Very Heavily Armored
19	108 in	84.8 in	66.3 in	9.6 in	456 lb-ft	5.74:1	50 in	1080 kWh	0.966 in	0.84 in
	Very Narrow	Short	Very Short Roof	Very Short Travel	Very Low Output	Very Short Gearing	Very Large Tire	Low Energy Capacity	Very Heavy-Duty Frame	Lightly Armored
20	102 in	92.3 in	76.1 in	8.6 in	715 lb-ft	4.79:1	45 in	990 kWh	0.733 in	0.28 in
	Very Narrow	Long	Moderate Height Roof	Very Short Travel	Very High Output	Short Gearing	Large Tire	Very Low Energy Capacity	Heavy-Duty Frame	Very Lightly Armored

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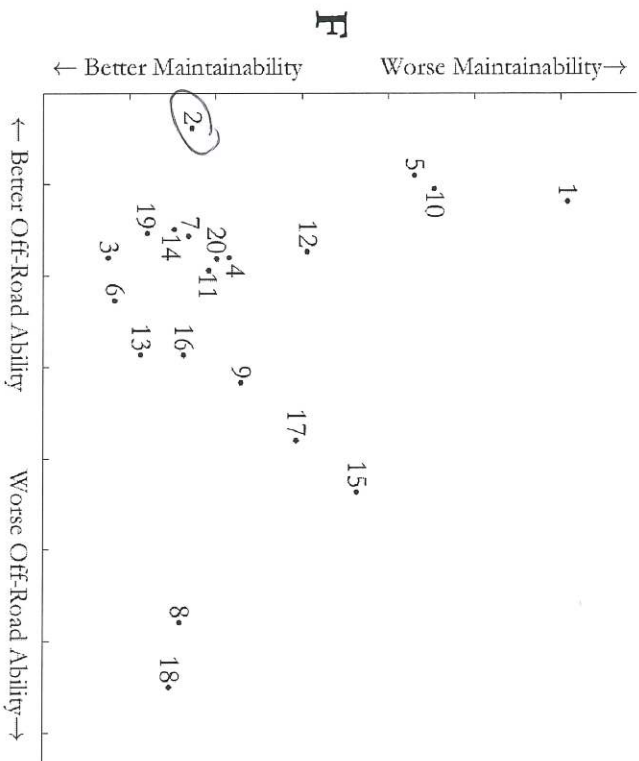
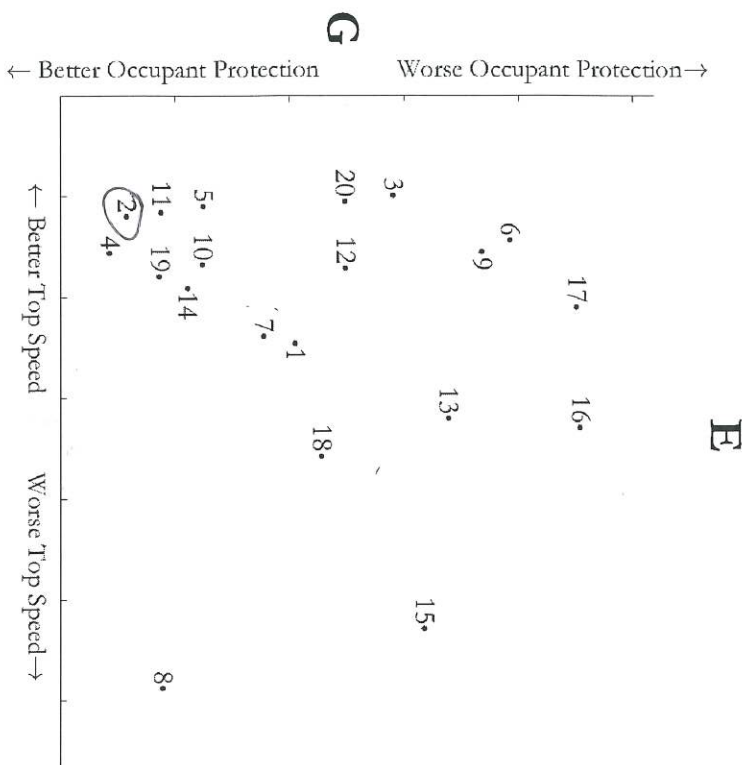
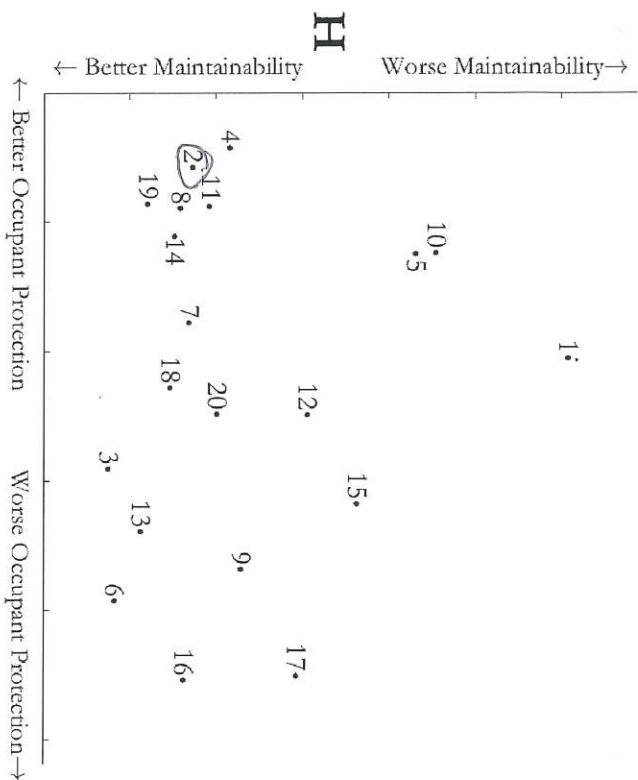
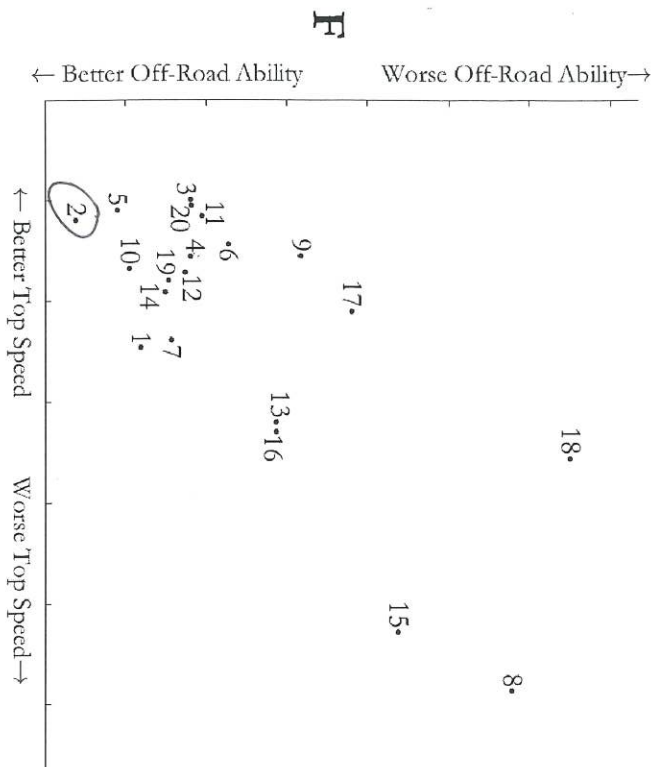
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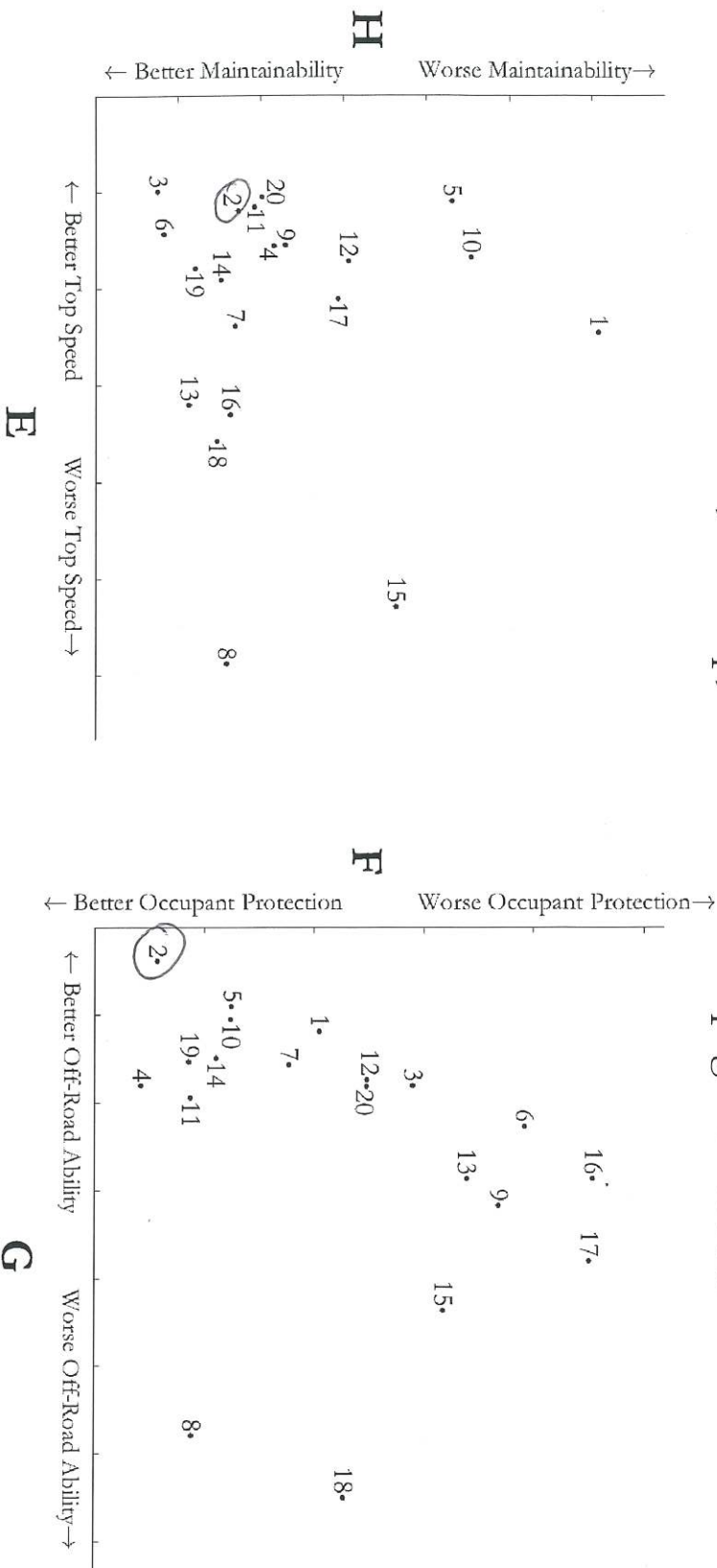
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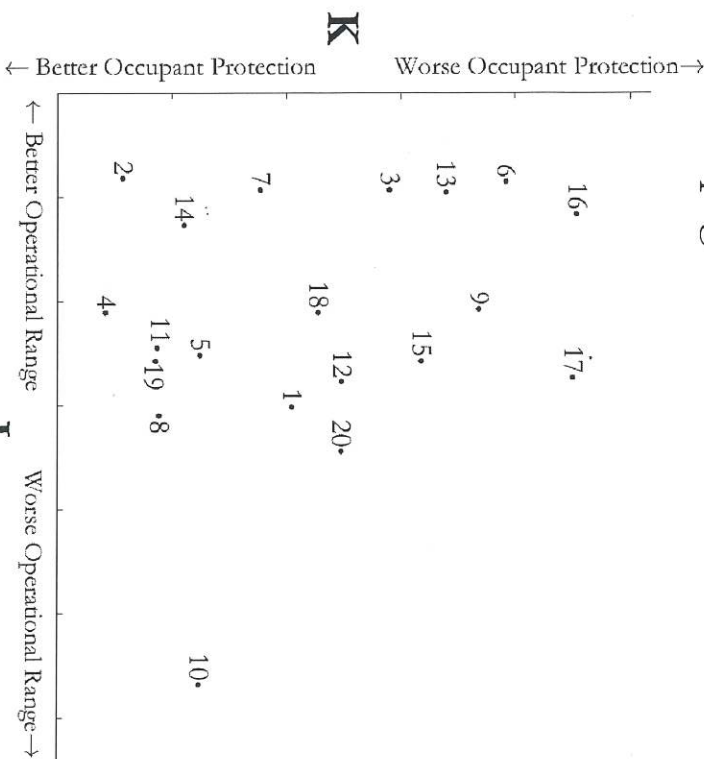
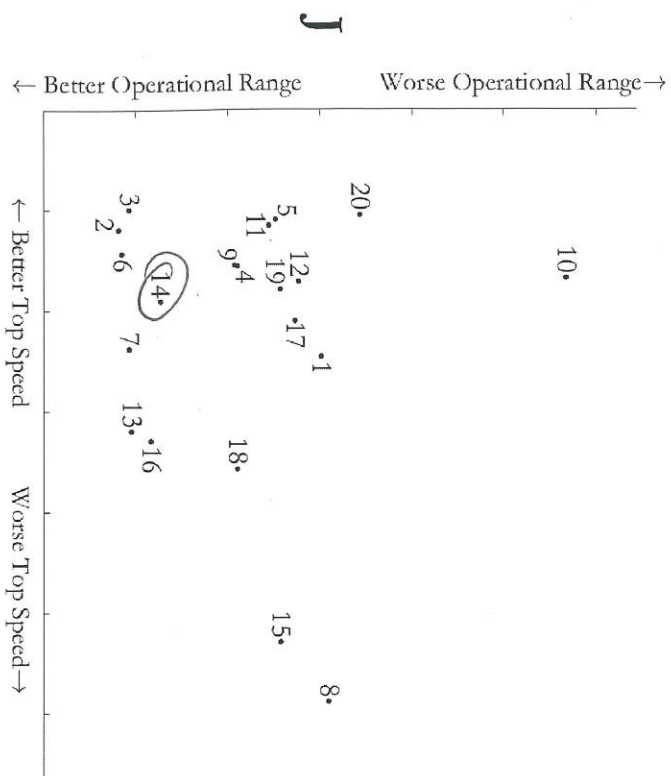
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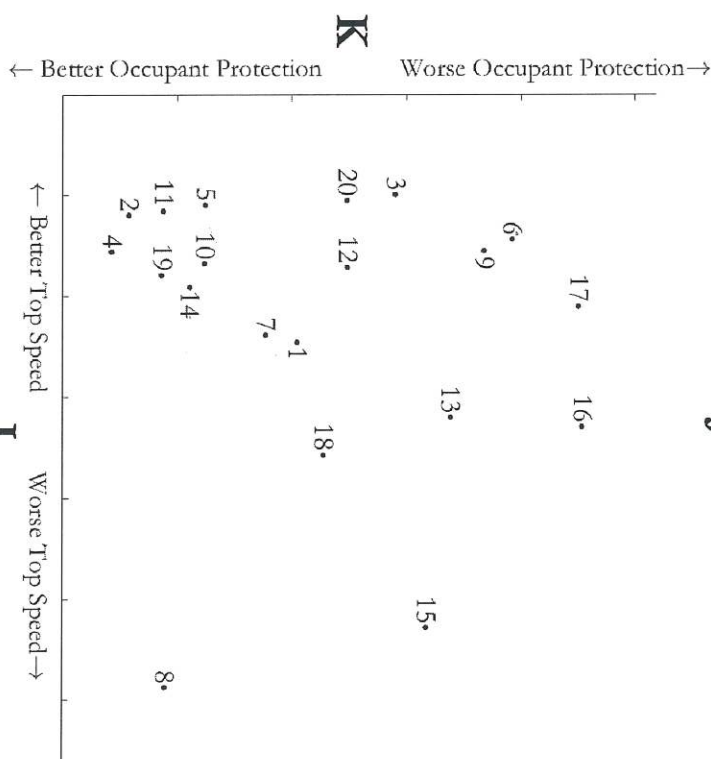
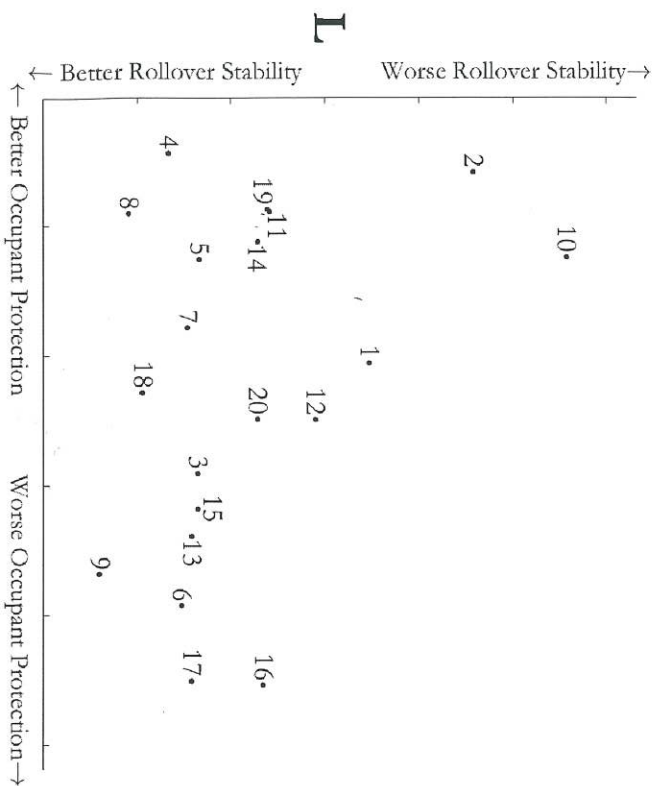
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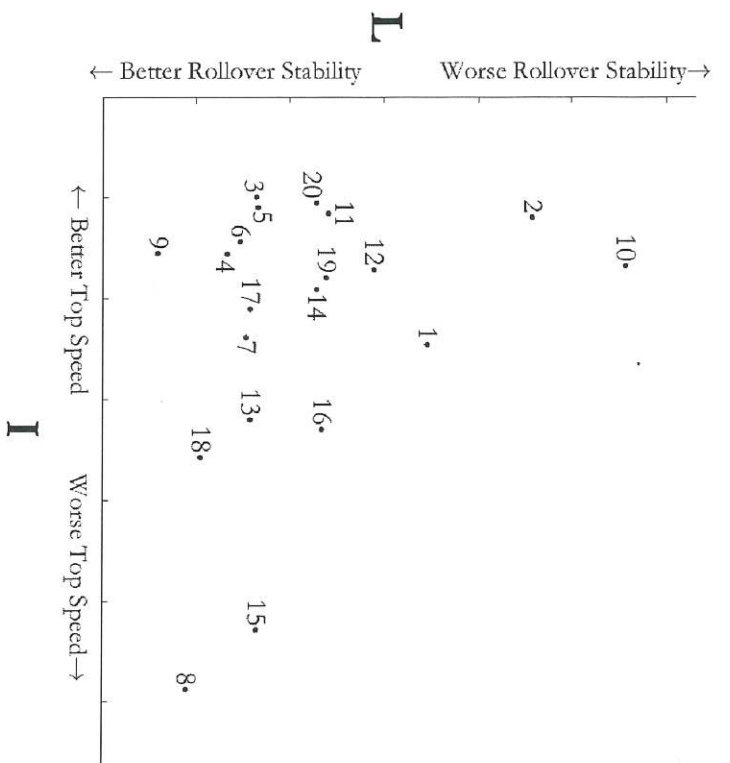
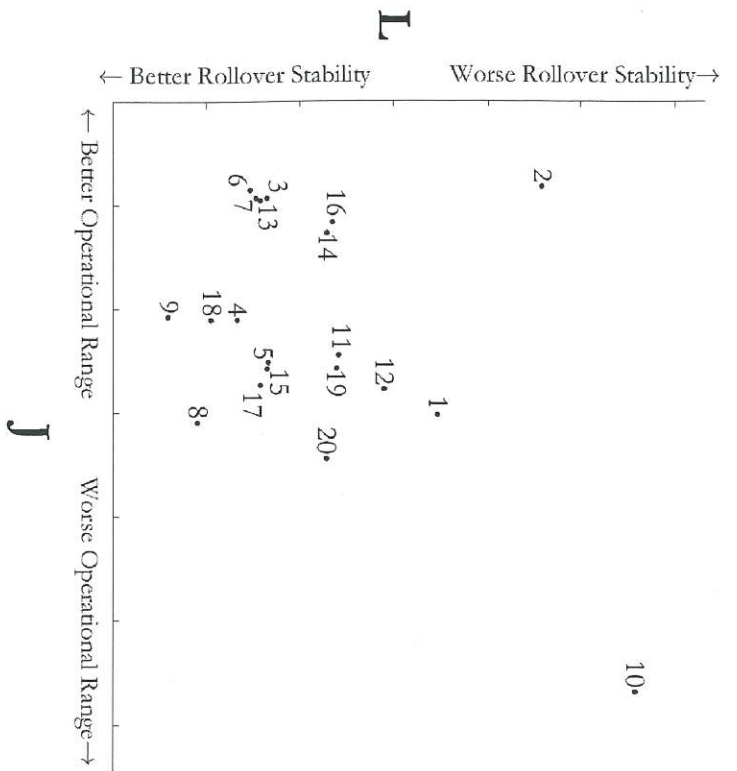
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